



# S.A. ENGINEERING COLLEGE

(Autonomous - Institute Level Research Centre)  
Sponsored by Dharma Naidu Educational and Charitable Trust  
Accredited by NBA, NAAC with 'A' Grade & ISO 9001 : 2015 Certified Institution  
Approved by A.I.C.T.E. and Affiliated to Anna University  
Poonamallee - Avadi Main Road, Thiruvarkadu, Chennai - 600 077.  
Phone : 2680 1999, 2680 1499, Fax : 2680 1899



**THIRU.Mr.S. AMARNAATH**  
**CORRESPONDENT**

Date: 13.07.2023

No.SAEC/Rec/2023-2024/Staff/Computer Science and Engineering Department

## APPOINTMENT ORDER

Sub: S.A.Engineering College, Chennai - 77-**Ms.P.Shobana Pritha**- Appointed  
as Visiting Faculty-**Computer Science and Engineering Department** -  
Orders -Issued..

Ref: Application dated 10.07.2023 of Ms.P.Shobana Pritha

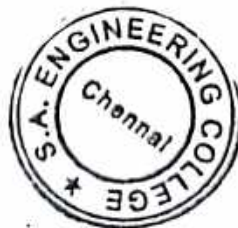
With reference to the application cited above and consequent to the interview attended on 10.07.2019, Ms.P.Shobana Pritha is appointed as **Visiting Faculty** in **Computer Science Engineering Department** of S.A.Engineering College, Chennai. Her Salary will be fixed as per college norms. She will also be eligible to draw other allowances as applicable from time to time.

She will have to submit all the original certificates to the Principal at the time of receiving the Intimation of Selection / Orders of Appointment. No TA/DA will be payable to her to join duty.

She will be on probation for a period of one year from the date of joining duty. She is eligible to draw the Salary as per the college norms during the probation period.

Other terms and conditions of employment will be as per the existing rules of the College. A copy of the conditions of employment is enclosed. She is requested to report for duty to the Principal on **24<sup>th</sup> July 2023.**

Also She is requested to submit his consent letter to the undersigned immediately, in confirmation of the receipt of the orders of appointment. Kindly acknowledge the receipt of the Appointment Order.



*S. Amarnaath*  
**CORRESPONDENT**

To  
Ms.P.Shobana Pritha  
Plot no:02 Phase-II, Garden City Extension  
Varadharajapuram, Manimangalam Road,  
Chennai-63

**S.A. ENGINEERING COLLEGE, CHENNAI - 77.**  
**CONDITIONS OF EMPLOYMENT**

01. You are required to carry out all the academic and other duties that may be assigned to you by the Principal of the College and the head of the department.
02. You will not undertake any job or tuition or any other activity in any manner, with or without remuneration, without the consent of the Management.
03. You will strive your best to uphold the name of the institution and aim at promoting the interests and welfare of the institution.
04. You will be eligible for one day casual leave for each month of completed service.
05. After completion of one year of service in the institution you will be eligible for vacation, subject to the condition that all the academic and other duties assigned are carried out before availing the vacation.
06. You will serve for a minimum period of two consecutive semesters i.e. a combination of successive odd and even semesters, or a combination of successive even and odd semesters. You will not be permitted to resign from the post in the middle of the academic year. Also you must serve in this Institution atleast two academic years without break.
07. For resignation during the probationary period, three month's notice in writing or payment of three month's salary in lieu thereof, is mandatory. Three month's notice shall be co-terminus with end of the semester/academic year. All the cases of resignation shall be subject to clause 6 above, as prerequisite.
08. For relieving staff at once due to sick or medical reason the staff member concerned should pay three months salary.
09. You will abide by other rules and regulations specified from time to time by the Principal/Management.
10. The Management reserves the right to terminate the services without notice on breach of discipline or conduct at any time during your service in this institution.
11. The above terms and conditions are subject to revision at the discretion of the Management from time to time.
12. Please sign the enclosed acceptance form for the offer of appointment and conditions of employment and mail the same to the Principal, S.A.Engineering College early.
13. TDS will be deducted from her salary as per rules in force.
14. PF/Medical Insurance/HRA/CCA/Professional Tax and other taxes which are applicable will be deducted from her/his salary periodically at the rate as prescribed by Government of Tamil Nadu.
15. As per the Undertaking given in the Proforma at the time of joining, those who have not completed two years are not eligible for break or prior notice for relieving.
16. One who availed one year break must appear before selection committee as a fresher for continuing in service.



*S. Annamalai*  
CORRESPONDENT

I P. Shobana Pritha have read all these Conditions given above and I agree to obey and serve in this Institution. I accept the offer of appointment as Visiting Faculty in CSE Department with the conditions stipulated by the S.A.ENGINEERING COLLEGE, CHENNAI. I will report for duty on 24/07/2023

DATE: 24/07/23

*P. Shobana*  
APPLICANT SIGNATURE



**VEI Technologies**  
We Design for future Technologies

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Date: 18-09-2025

**EXPERIENCE CERTIFICATE**

4/432 Muthamil Nagar,  
Mudichur Road  
Old Perungalathur,  
Chennai – 600048  
[shobanaveitech@gmail.com](mailto:shobanaveitech@gmail.com)  
PH: 7358426805

**TO WHOMSOEVER IT MAY CONCERN**

This is to certify that **Ms. P. Shobana Pritha** was employed with **VEI Technologies Pvt Ltd**, Chennai, as an **HR & Technical Manager** from **2/6/2015** to **Till date**, serving the organization with dedication for a period of **15 years**.

During her tenure with us, **Ms. P. Shobana Pritha** displayed exemplary leadership and professionalism in handling both human resources and technical management functions. Her key responsibilities included:

- Overseeing recruitment, employee relations, and HR compliance
- Managing technical teams and ensuring successful project delivery
- Implementing HR policies and fostering a productive work culture
- Collaborating with cross-functional teams to meet business objectives
- Mentoring team members and promoting skill development

**Ms. P. Shobana Pritha's** contributions significantly benefited the organization's growth and operational excellence. Her ability to manage both human resources and technical aspects with equal efficiency was highly commendable.

We thank her for her valuable service and wish her continued success in all her future endeavors.

**Regards,**  
Dr.B.Ezhilavan  
CEO & FOUNDER  
VEI TECHNOLOGIES PVT LTD  
CHENNAI



## IDENTITY CARD



EMPLOYEE ID  
VEI 007

  
AB+

**P.SHOBANA PRITHA, MCA., M.Phil. Phd.,**  
SOFTWARE DEVELOPER & TRAINER





## S. A. ENGINEERING COLLEGE, CHENNAI-77

(Autonomous-Institute Level Research Centre-Affiliated to Anna University)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution  
Poonamallee – Avadi Main Road, Thiruvirkadu, Chennai – 600 077.

### DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR: 2025-2026

YEAR:III

#### COURSE SYLLABUS

#### FULL STACK DEVELOPMENT

##### Course Objectives (COs)

By the end of this course, students will be able to:

- Understand the fundamentals of web technologies, including HTML, CSS, and JavaScript, and develop responsive and interactive web pages.
- Learn and implement modern front-end frameworks (React) to build dynamic single-page applications with reusable components.
- Develop server-side applications using Node.js and Express.js, design RESTful APIs, and handle client-server communication efficiently.
- Gain proficiency in working with relational and non-relational databases, including CRUD operations, data modelling, and integration with backend servers.
- Integrate front-end, backend, and database into a full-stack application, perform testing and debugging, and deploy projects on cloud platforms.

#### Module 1: Web Foundations

11

Introduction to Web Development – Client-Server Architecture, HTTP, APIs-HTML5-Structure of a web Page-Forms, multimedia, semantic tags-CSS3-Selectors, Box model, Flexbox, Grid, Responsive design, animations, Transitions-JavaScript (ES6)-Variables, data types, loops, functions-DOM manipulation and events-JSON and Fetch API,

#### Module 2: Front-End Framework

11

Introduction to React (or Angular/Vue – as per institution)-React Components, Props, and State- Hooks (useState, useEffect)- Conditional Rendering and Lists- Event Handling and Forms in React- Routing (React Router)- Integration with REST APIs

#### Module-3 Backend Development with Node.js

11

Introduction to Node.js and Express.js- Setting up Express server- Middleware, Routing, and REST API design- Handling requests and responses- Working with JSON and API testing using Postman- Introduction to authentication (JWT / Session)

#### Module-4 Database and Data Handling

11

Relational Database (MySQL / PostgreSQL) – CRUD operations- Introduction to NoSQL -MongoDB)- Connecting Node.js with Database (Mongoose / Sequelize)- Schema design and data modelling- Querying and aggregation- Data validation and error handling

#### Module-5 Integration, Deployment & Project

10

Integrating Front-end (React) with Backend (Node.js + DB)- Authentication and Authorization- Environment variables and configuration- Deployment – GitHub, Render, Vercel, or AWS- Version Control (Git & GitHub basics)- Testing and Debugging- Project Demo and Evaluation

Total No.of.Hours: 54

# Course Outcomes (COs)

After completing the course, students will be able to:

**CO1:** Design and implement responsive, user-friendly web pages using HTML, CSS, and JavaScript.

**CO2:** Build dynamic, component-based single-page applications using React (or chosen front-end framework).

**CO3:** Develop and test server-side RESTful APIs using Node.js and Express.js to handle real-world client requests.

**CO4:** Implement data persistence using relational (MySQL/PostgreSQL) and non-relational (MongoDB) databases and connect them with backend services.

**CO5:** Integrate front-end, backend, and database into a functional full-stack application, and deploy it on cloud platforms while following industry-standard best practices.

| COs | K-Map | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | K3    | 3   | 2   | 2   | 1   | 3   | 1   | 1   | 1   | 2   | 2    | 1    | 2    | 1    |      |
| CO2 | K3    | 3   | 3   |     | 2   | 3   |     | 1   | 1   |     |      |      |      | 1    |      |
| CO3 | K3    | 3   | 3   |     |     |     |     | 1   | 1   |     |      |      |      | 1    |      |
| CO4 | K3    | 3   |     | 3   | 2   |     | 1   |     | 1   | 3   | 2    |      | 2    |      | 1    |
| CO5 | K3    | 3   | 3   | 3   | 2   | 3   |     |     |     |     | 3    | 3    | 3    |      | 1    |



FACULTY INCHARGE



AEC COORDINATOR



HOD/CSE

Full Stack Development - 60 MCQs  
Academic year (2025-2026)  
(M1-M3)

Answers follow each question.

\* Indicates required question

1. NAME

2. REGISTER NUMBER

3. SECTION

Mark only one oval.

- ☐ A  
☐ B  
☐ C

4. 1. Which tag is used for embedding a video? \*

Mark only one oval.

- ☐ a) <media>  
☐ b) <embed>  
☐ c) <video>  
☐ d) <source>

5. 2. Which HTML5 tag is used for defining independent, self-contained content?

Mark only one oval.

- ☐ a) <section>  
☐ b) <article>  
☐ c) <div>  
☐ d) <aside>

6. 3. Which CSS layout technique allows both rows and columns control?

Mark only one oval.

- ☐ a) Flexbox  
☐ b) Grid  
☐ c) Float  
☐ d) Inline-block

7. 4. What is the correct order of the CSS box model?

Mark only one oval.

- ☐ a) Content → Padding → Border → Margin  
☐ b) Margin → Border → Padding → Content  
☐ c) Padding → Border → Margin → Content  
☐ d) Content → Border → Padding → Margin

8. 5. Which HTTP status code indicates a successful GET request?

Mark only one oval.

- ☐ a) 200 OK  
☐ b) 301 Moved Permanently  
☐ c) 404 Not Found  
☐ d) 500 Internal Server Error

9. Which JavaScript method converts a JSON string into an object?

Mark only one oval.

- ☐ a) JSON.parse()  
☐ b) JSON.stringify()  
☐ c) JSON.toObject()  
☐ d) JSON.convert()

10. 7. In HTML forms, which attribute specifies where to send form data?

Mark only one oval.

- ☐ a) target  
☐ b) action  
☐ c) method  
☐ d) enctype

11. 8. Which JavaScript method converts a JSON string into an object?

Mark only one oval.

- ☐ a) JSON.parse()  
☐ b) JSON.stringify()  
☐ c) JSON.toObject()  
☐ d) JSON.convert()

12. 9. Which selector has the highest CSS specificity?

Mark only one oval.

- ☐ a) Element selector  
☐ b) Class selector  
☐ c) ID selector  
☐ d) Universal selector

13. 10. Which HTML5 element is used to embed audio files?

Mark only one oval.

- ☐ a) <media>  
☐ b) <audio>  
☐ c) <sound>  
☐ d) <music>

14. 11. Which JavaScript keyword declares a block-scoped variable?

Mark only one oval.

- ☐ a) var  
☐ b) let  
☐ c) const  
☐ d) both b and c

15. 12. Which HTML tag defines important text semantically? What does the Fetch API return?

Mark only one oval.

- ☐ a) <em>  
☐ b) <strong>  
☐ c) <b>  
☐ d) <i>

16. 13. Which property makes a website responsive?

Mark only one oval.

- ☐ a) max-width  
☐ b) position  
☐ c) float  
☐ d) display



17. 14. Which event occurs when a user clicks an HTML element?

Mark only one oval.

- ☐ a) onmouseover  
☐ b) onload  
☐ c) onclick  
☐ d) onchange

18. 15. What does the Fetch API return?

Mark only one oval.

- ☐ a) A Promise  
☐ b) A JSON object  
☐ c) An XMLHttpRequest  
☐ d) A Callback

19. 16. Which CSS property adds shadow to text?

Mark only one oval.

- ☐ a) text-style  
☐ b) font-shadow  
☐ c) text-shadow  
☐ d) shadow

20. 17. Which HTTP method is idempotent?

Mark only one oval.

- ☐ a) POST  
☐ b) GET  
☐ c) PATCH  
☐ d) PUT

21. 18. Which API allows asynchronous communication between client and server?

Mark only one oval.

- ☐ a) WebSocket  
☐ b) Fetch API  
☐ c) REST API  
☐ d) GraphQL

22. 19. What is DOM in web development?

Mark only one oval.

- ☐ a) Document Object Model  
☐ b) Data Object Manager  
☐ c) Display Order Model  
☐ d) Dynamic Object Mapping

23. 20. Which HTML5 element is used to embed audio files? \*

Mark only one oval.

- ☐ a) <media>  
☐ b) <audio>  
☐ c) <sound>  
☐ d) <music>

24. 21. Which method is used to create components in modern React?

Mark only one oval.

- ☐ a) React.createComponent()  
☐ b) React.Component()  
☐ c) Functional components using hooks  
☐ d) class components only

25. 22. Which hook is used for managing component state?

Mark only one oval.

- ☐ a) useEffect
- ☐ b) useState
- ☐ c) useReducer
- ☐ d) useMemo

26. 23. What is the virtual DOM?

Mark only one oval.

- ☐ a) An actual browser DOM
- ☐ b) A lightweight copy of the DOM in memory
- ☐ c) A server-side DOM
- ☐ d) A CSS object model

27. 24. Which hook replaces component did mount?

Mark only one oval.

- ☐ a) useEffect
- ☐ b) useState
- ☐ c) useRef
- ☐ d) useMemo

28. 25. Which package is used for navigation in React?

Mark only one oval.

- ☐ a) react-dom
- ☐ b) react-router-dom
- ☐ c) react-path
- ☐ d) react-navigator

29. 26. Props in React are:

Mark only one oval.

- ☐ a) Mutable data
- ☐ b) Read-only data
- ☐ c) Functions only
- ☐ d) Used only in class components

30. 27. How do you handle forms in React?

Mark only one oval.

- ☐ a) Using document.getElementById()
- ☐ b) With controlled components
- ☐ c) Using querySelector()
- ☐ d) With CSS selectors

31. 28. What is JSX?

Mark only one oval.

- ☐ a) A function in React
- ☐ b) A syntax extension combining JavaScript and HTML
- ☐ c) A JSON file
- ☐ d) A CSS preprocessor

32. 29. Which lifecycle method is used to clean up effects?

Mark only one oval.

- ☐ a) componentWillUnmount
- ☐ b) componentDidMount
- ☐ c) componentDidUpdate
- ☐ d) componentShouldUpdate

33. 30. Which hook is used to perform side effects?

Mark only one oval.

- ☐ a) useState  
☐ b) useRef  
☐ c) useEffect  
☐ d) useCallback

34. 31. What is key prop used for in lists?

Mark only one oval.

- ☐ a) Uniquely identify list items  
☐ b) Style list items  
☐ c) Track re-render count  
☐ d) To make list dynamic

35. 32. How can you prevent component re-render?

Mark only one oval.

- ☐ a) Using React.memo  
☐ b) Using useEffect  
☐ c) Using useCallback  
☐ d) Using useReducer

36. 33. What is the purpose of useRef?

Mark only one oval.

- ☐ a) Track state changes  
☐ b) Access DOM elements  
☐ c) Define styles  
☐ d) Bind methods

37. 34. Which file is typically the root component in a React app?

Mark only one oval.

- ☐ a) index.js  
☐ b) App.js  
☐ c) main.js  
☐ d) root.js

38. 35. How can data be passed from parent to child?

Mark only one oval.

- ☐ a) Using hooks  
☐ b) Through props  
☐ c) Via state  
☐ d) With context only

39. 36. Which tool is used to create a React app quickly?

Mark only one oval.

- ☐ a) npm init  
☐ b) npx create-react-app  
☐ c) yarn new react  
☐ d) react-cli

40. 37. React uses which rendering model?

Mark only one oval.

- ☐ a) Server-side rendering only  
☐ b) Client-side rendering  
☐ c) Hybrid rendering  
☐ d) Batch rendering



41. 38. Which package manager is used with React?

Mark only one oval.

- ☐ a) composer
- ☐ b) pip
- ☐ c) npm
- ☐ d) gem

42. 39. Which hook helps optimize performance by memoizing values?

Mark only one oval.

- ☐ a) useMemo
- ☐ b) useEffect
- ☐ c) useReducer
- ☐ d) useState

43. 40. Which context API is used for global state?

Mark only one oval.

- ☐ a) useState
- ☐ b) createContext
- ☐ c) useRef
- ☐ d) useReducer

44. 41. Which command initializes a Node.js project?

Mark only one oval.

- ☐ a) npm init
- ☐ b) node start
- ☐ c) npm install
- ☐ d) node init

45. 42. Which module is used to create a server in Node.js?

Mark only one oval.

- ☐ a) http
- ☐ b) fs
- ☐ c) url
- ☐ d) path

46. 43. What is Express.js?

Mark only one oval.

- ☐ a) A template engine
- ☐ b) A Node.js web framework
- ☐ c) A database
- ☐ d) A CSS library

47. 44. Which method handles GET requests in Express?

Mark only one oval.

- ☐ a) app.fetch()
- ☐ b) app.route()
- ☐ c) app.get()
- ☐ d) app.send()

48. 45. Middleware functions in Express are used for:

Mark only one oval.

- ☐ a) Handling database
- ☐ b) Handling requests and responses
- ☐ c) Defining schemas
- ☐ d) Routing only

49. 46. Which package is used for API testing?

Mark only one oval.

- ☐ a) Jest  
☐ b) Mocha  
☐ c) Postman  
☐ d) Supertest

50. 47. What does `res.send()` do?

Mark only one oval.

- ☐ a) Sends a file  
☐ b) Sends an HTTP response  
☐ c) Sends JSON data  
☐ d) Ends the connection

51. 48. Which module handles file operations?

Mark only one oval.

- ☐ a) fs  
☐ b) os  
☐ c) http  
☐ d) url

52. 49. How do you import an external module?

Mark only one oval.

- ☐ a) include()  
☐ b) require()  
☐ c) importlib()  
☐ d) load()

53. 50. Which HTTP method updates existing data?

Mark only one oval.

- ☐ a) POST  
☐ b) GET  
☐ c) PUT  
☐ d) DELETE

54. 51. What is JWT used for?

Mark only one oval.

- ☐ a) Database connection  
☐ b) Authentication  
☐ c) Data encryption  
☐ d) Session storage

55. 52. Which object represents the HTTP request?

Mark only one oval.

- ☐ a) req  
☐ b) res  
☐ c) app  
☐ d) next

56. 53. Which Express middleware parses incoming JSON?

Mark only one oval.

- ☐ a) express.static  
☐ b) express.json()  
☐ c) body-parser  
☐ d) urlencoded

57. 54. What port is commonly used by Node.js apps? \*

Mark only one oval.

- ☐ a) 80  
☐ b) 3000  
☐ c) 8080  
☐ d) 443

58. 55. Which command runs a Node app?

Mark only one oval.

- ☐ a) npm start  
☐ b) node app.js  
☐ c) npm install  
☐ d) run node

59. 56. Which environment variable stores the running port? \*

Mark only one oval.

- ☐ a) NODE\_ENV  
☐ b) PORT  
☐ c) HOST  
☐ d) ENV

60. 57. Which tool helps restart server automatically? \*

Mark only one oval.

- ☐ a) Nodemon  
☐ b) PM2  
☐ c) Forever  
☐ d) All of the above

61. 57. Which method sends a JSON response? \*

Mark only one oval.

- ☐ a) res.json()  
☐ b) res.data()  
☐ c) res.sendJSON()  
☐ d) res.output()

62. 58. What does REST stand for?

Mark only one oval.

- ☐ a) Representational State Transfer  
☐ b) Remote Server Testing  
☐ c) Runtime Environment State  
☐ d) Resource Exchange Syntax

63. 59. Which Express method defines a route parameter? \*

Mark only one oval.

- ☐ a) app.param()  
☐ b) app.route()  
☐ c) app.link()  
☐ d) app.listen()

64. 60. Which tool helps restart server automatically?

Mark only one oval.

- ☐ a) Nodemon  
☐ b) PM2  
☐ c) Forever  
☐ d) All of the above




**S.A. ENGINEERING COLLEGE, CHENNAI-77**  
 (Autonomous-Institute Level Research Centre - Affiliated to Anna University)  
 DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

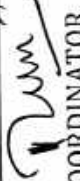
**ACTION TAKEN REPORT BASED ON STUDENT FEEDBACK**

ACADEMIC YEAR: 2025-2026

SUBJECT: FULL STACK DEVELOPMENT

| Feedback Category                        | Summary of Student Feedback                                                      | Action Taken                                                                                                                       |
|------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Guest Speaker Knowledge & Presentation   | Majority rated 4.0-5.0, indicating strong delivery and subject expertise.        | Continue inviting domain experts with engaging presentation styles.                                                                |
| Relevance of Topic                       | High relevance (avg. rating :4.5) to students' academic and career interests.    | Maintain alignment with curriculum and emerging technologies.                                                                      |
| Q&A Effectiveness                        | Most students felt their queries were addressed (avg. rating: 4.2).              | Encourage interactive Q&A sessions and allocate more time for discussion.                                                          |
| Engineering Fundamentals (PO1)           | Strong enhancement reported (avg. rating :4.5).                                  | Reinforce foundational concepts through practical examples.                                                                        |
| Problem Analysis (PO2)                   | Positive impact noted (avg. rating :4.3).                                        | Include more case studies and real-world problem breakdowns.                                                                       |
| Design & Implementation (PO3)            | Students appreciated design-oriented learning (avg. rating :4.2).                | Introduce mini design challenges during sessions.                                                                                  |
| Investigation & Data Analysis (PO4)      | Good engagement with analytical tasks (avg. rating :4.3).                        | Provide structured data sets and guided analysis exercises.                                                                        |
| Use of Modern Tools (PO5)                | High appreciation for tool exposure (avg. rating :4.4).                          | Continue integrating tools like ML, IoT, and cloud platforms.                                                                      |
| Communication Skills (PO10)              | Moderate improvement noted (avg. rating :4.2).                                   | Include short presentations or peer discussions.                                                                                   |
| Lifelong Learning Awareness (PO12)       | Strong awareness raised (avg. rating :4.4).                                      | Emphasize evolving tech trends and self-learning resources.                                                                        |
| Application of Math & Computation (PSO1) | Students recognized real-world relevance (avg. rating :4.3).                     | Include more examples linking theory to engineering applications.                                                                  |
| Domain Knowledge Application (PSO2)      | High ratings for practical integration (avg. rating :4.4).                       | Encourage project-based learning with domain-specific tools.                                                                       |
| Suggestions for Improvement              | Common themes: "Need air conditioner", "Need more sessions", "More informative". | - Improve venue comfort (AC installation planned).<br>- Schedule follow-up sessions.<br>- Enhance content depth and interactivity. |

  
 CLASS ADVISOR

  
 YEAR COORDINATOR

  
 HoD/CSE



**S. A. ENGINEERING COLLEGE, CHENNAI-77**

*(An Autonomous Institution Affiliated to Anna University, Chennai)*

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Poonamallee – Avadi Main Road, Thiruverkadu, Chennai – 600 077.*

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ACADEMIC YEAR: 2024-2025**

**YEAR:III**

**COURSE SYLLABUS**

**UI/UX DESIGN**

Course Objectives (COs)

By the end of this course, students will be able to:

- Introduce the fundamental principles of User Interface (UI) and User Experience (UX) design.
- Develop skills in user research, persona creation, and understanding user needs and behavior.
- Enable students to design effective information architecture, wireframes, and prototypes.
- Familiarize students with industry-standard design tools for interface creation and usability testing.
- Empower students to create, evaluate, and refine interactive design systems following usability principles and accessibility standards.

**Module 1: Introduction to UI/UX Design**

11

Overview of UI/UX Design-Difference between UI and UX-Design Thinking Process and Human-Centered Design-Human-Computer Interaction (HCI) Basics-Visual Design Principles – Balance, Contrast, Alignment, Hierarchy-Color Theory, Typography, and Layout Design-Case Study: Evaluating existing website interfaces

**Module 2: User Research and Requirement Analysis**

11

Introduction to User Research-Types of User Research: Qualitative and Quantitative-Data Collection Methods: Interviews, Surveys, Observations-Empathy Mapping and Persona Creation-User Journey Mapping and Scenario Building-Defining Design Goals and Problem Statements

**Module 3: Information Architecture and Wireframing**

11

Information Architecture – Concepts and Importance-Content Hierarchy and Navigation Design-Creating Sitemaps and Flowcharts-Wireframing Techniques – Low-fidelity & High-fidelity-Layout Design and Visual Hierarchy-Tools for Wireframing (Figma, Adobe XD, Sketch)

**Module 4: Prototyping and Visual Design**

11

Introduction to Prototyping-Interactive Components and Transitions-Style Guides, Design Systems, and UI Kits-Color Schemes, Typography, and Iconography-Responsive Design Principles (Mobile, Tablet, Desktop)-Accessibility and Inclusive Design

**Module 5: Usability Testing and Project Development**

10

Usability Testing Concepts and Methods-Heuristic Evaluation and Cognitive Walkthrough-Accessibility Standards (WCAG)-Collecting and Analysing User Feedback-Design Iteration and Refinement-Final Project Presentation and Review

**Total No.of.Hours: 54**



# Course Outcomes (COs)

After completing the course, students will be able to:

**CO1:** Explain the principles and process of UI/UX design and its role in digital product development.

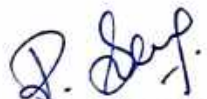
**CO2:** Conduct user research and create user personas, empathy maps, and journey maps to understand user requirements.

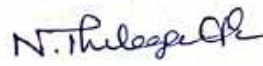
**CO3:** Design user interfaces through wireframes, information architecture, and prototypes using standard design practices.

**CO4:** Apply usability testing techniques to evaluate and improve interface designs using tools such as Figma, Adobe XD, or Sketch.

**CO5:** Integrate aesthetics, accessibility, and interaction design principles to create responsive and user-centered digital experiences.

| COs | K-Map | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | K2    | 3   | 2   | 2   | 1   | 3   | 1   | 1   | 1   | 2   | 1    | 1    | 2    | 1    |      |
| CO2 | K3    | 3   | 2   |     |     |     |     |     |     | 2   |      |      |      | 1    |      |
| CO3 | K3    | 3   | 2   |     |     |     |     |     |     | 2   |      |      |      | 1    |      |
| CO4 | K3    | 3   | 1   | 3   | 2   |     | 1   |     |     | 2   |      |      | 2    |      | 1    |
| CO5 | K4    | 3   | 3   | 3   | 2   | 3   |     |     |     | 2   | 2    | 2    | 2    |      | 1    |

  
FACULTY INCHARGE

  
AEC COORDINATOR

  
HOD/CSE